

Consider a location family given by a density f_μ parameterized by $\mu \in \mathbb{R}$, given by

$$f_\mu(x) = \frac{1}{2} e^{-|x-\mu|}, x \in \mathbb{R}.$$

This is known as the [Laplace](#) or Double Exponential family with location parameter μ .

Given an i.i.d. random sample $X_1, X_2, \dots, X_n$ from f_μ , where μ is unknown, we are interested in obtaining a 95% confidence interval for μ .

✓ Approximate Normal Interval

It is easy to verify that $E(X_1) = \mu$ and $Var(X_1) = 2$. Let $\bar{X}$ be the sample mean of $X_1, X_2, \dots, X_n$. Then, by the Central Limit Theorem, we know that the approximate distribution of $\bar{X} - \mu$ is

$$\bar{X} - \mu \approx N(0, 2/n).$$

In other words, since $P(-1.96 < N(0, 1) < 1.96) \approx 0.95$, we have

$$P(1.96\sqrt{2/n} < \bar{X} - \mu < 1.96\sqrt{2/n}) \approx 0.95 \implies P(\bar{X} - 1.96\sqrt{2/n} < \mu < \bar{X} + 1.96\sqrt{2/n}) \approx 0.95$$

This leads to the following function to obtain an approximate 95% confidence interval for μ .

```
laplace_ci_mean <- function(x)
{
  n <- length(x)
  xbar <- mean(x)
  c(lower = xbar - 1.96 * sqrt(2/n), upper = xbar + 1.96 * sqrt(2/n))
}
```

We can simulate such intervals many times to estimate two relevant characteristics: the coverage probability (which should be approximately 95%) and the average length (shorter intervals are more useful). Both of these can depend on the sample size n (the Normal approximation may not be).

We can simulate observations from the Laplace distribution by first noting that the CDF F is given by

$$F_\mu(x) = \begin{cases} \frac{1}{2}e^{x-\mu} & \text{if } x < \mu \\ 1 - \frac{1}{2}e^{-(x-\mu)} & \text{if } x \geq \mu \end{cases},$$

and then recall that if $U \sim U(0, 1)$, then $F_\mu^{-1}(U)$ has CDF F_μ . Here,

$$F_\mu^{-1}(u) = \begin{cases} \mu + \log 2u & \text{if } u < \frac{1}{2} \\ \mu - \log 2(1-u) & \text{if } u \geq \frac{1}{2} \end{cases}.$$

```
rlaplace <- function(n, mu = 0)
{
  u <- runif(n)
  ifelse(u < 0.5, mu + log(2*u), mu - log(2 * (1-u)))
}
```

We can now use this to check whether the target coverage probability is approximately satisfied.

```
cimean51 <- replicate(10000, laplace_ci_mean(rlaplace(51, mu = 5))) |> t() |> as.data.frame()
cimean501 <- replicate(10000, laplace_ci_mean(rlaplace(501, mu = 5))) |> t() |> as.data.frame()
print(head(cimean51))
```

```

#>      lower  upper
#> 1 4.322367 5.098642
#> 2 4.237200 5.013476
#> 3 4.232194 5.008469
#> 4 4.477667 5.253943
#> 5 4.585784 5.362060
#> 6 4.806288 5.582564
```

```
with(cimean51, sum(lower < 5 & upper > 5)) / nrow(cimean51)
```

```
#> 0.9527
```

```
with(cimean501, sum(lower < 5 & upper > 5)) / nrow(cimean501)
```

```
↗ 0.9518
```

We can also compute the average length of the confidence intervals as follows.

```
print(with(cimean51, mean(upper - lower)))
print(with(cimean501, mean(upper - lower)))
```

```
↗ [1] 0.7762757
  [1] 0.247675
```

✓ Your assignment

The confidence interval outlined above is based on the sample mean, using the fact that the function $\bar{X} - \mu$ is a *pivotal quantity* whose distribution is approximately known.

Your task is to obtain a similar *exact* confidence interval based on the sample median $\tilde{X}$. It is intuitively clear that $\tilde{X} - \mu$ is also a pivotal quantity, whose distribution does not depend on μ . If we can find a constant c such that

$$P(-c < \tilde{X} - \mu < c) = 0.95,$$

then we can repeat the process above to obtain a 95% confidence interval for μ . To obtain the value of c , we need to find the distribution of $\tilde{X} - \mu$. You can do so using the following results:

Result 1: If $U_1, \dots, U_n \sim \text{i.i.d. } U(0, 1)$, where $n = 2k + 1$ is odd, then the sample median $\tilde{U} \sim \text{Beta}(k, k)$.

Result 2: By invariance of the median under monotone transformations, $F_\mu^{-1}(\tilde{U})$ and $\tilde{X}$ have the same distribution.

Instructions

Using the results above, compute c_n as a function of n such that

$$P(-c_n < \tilde{X} - \mu < c_n) = 0.95,$$

for n odd. Use it to write a function named `laplace_ci_median` to compute a confidence interval for μ .

```
laplace_ci_median <- function(x)
{
  n <- length(x)
  xmed <- median(x)
  # begin your code here
  c_n <- ...
  # end your code here
  c(lower = xmed - c_n, upper = xmed + c_n)
}
```

Once you are satisfied with your solution, run the following cells to verify the coverage probability and average length of the resulting confidence intervals.

```
cimed51 <- replicate(10000, laplace_ci_median(rlaplace(51, mu = 5))) |> t() |> as.data.frame()
cimed501 <- replicate(10000, laplace_ci_median(rlaplace(501, mu = 5))) |> t() |> as.data.frame()
print(with(cimed51, sum(lower < 5 & upper > 5) / nrow(cimed51)))
print(with(cimed501, sum(lower < 5 & upper > 5) / nrow(cimed501)))
print(with(cimed51, mean(upper - lower)))
print(with(cimed501, mean(upper - lower)))
```

Based on the above results, which confidence interval would you prefer?

Instructions for submission will be provided in due course.